

iCal Calendar Synchronization

A complete specification for two-way availability synchronization between our travel booking platform and third-party distribution channels.

MODULE	Calendar Sync (iCal Import and Export)
AUDIENCE	UI/UX Design, Frontend, Backend, QA
SOURCE OF TRUTH	Our own booking platform
STATUS	Ready for design and implementation
DATE	01 August 2026

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1. Feature Overview

1.1 What iCal is

iCal (RFC 5545, file extension `.ics`) is a plain text calendar format. A `.ics` feed is a URL that returns a text document containing a list of `VEVENT` blocks. Each block describes an occupied date range with a start date, an end date, a unique identifier and an optional summary.

A minimal event looks like this:

```
BEGIN:VEVENT
UID:booking-48219@ourplatform.com
DTSTART;VALUE=DATE:20260814
DTEND;VALUE=DATE:20260818
SUMMARY:Reserved
END:VEVENT
```

Every major OTA (Airbnb, Booking.com, Expedia, Vrbo, Viator, GetYourGuide, MakeMyTrip) can publish and consume this format. It is the lowest common denominator for availability exchange.

1.2 What exporting means

We publish a read-only, token-protected URL for each trip or listing. It returns a `.ics` feed of every date the trip is unavailable, including bookings made on our platform and manual blocks placed by the operator.

The operator pastes this URL into Airbnb, Booking.com and other channels. Those platforms poll the URL on their own schedule and block the matching dates in their own inventory.

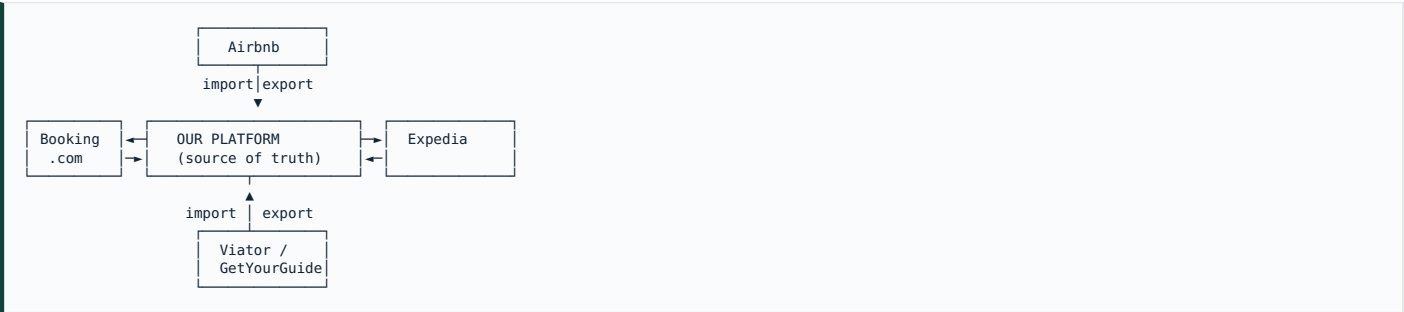
Direction: Our platform → OTA.

1.3 What importing means

The operator copies an iCal URL from each OTA and adds it to our platform. Our scheduler downloads each feed at a configured interval, parses the events, and writes the occupied dates into our availability layer as blocked dates attributed to that source.

Direction: OTA → our platform.

Import and export together produce a hub-and-spoke topology. Our platform is the hub. Each channel connects to the hub in both directions. Channels do not need to talk to each other.



1.4 Why users need it

A travel operator who lists the same inventory on four channels has four independent availability calendars. Without synchronization, two guests can book the same date on two different platforms. The operator then has to cancel one, which triggers penalties, ranking damage and refund handling on most OTAs.

Calendar sync removes the manual reconciliation work and reduces the double-booking window from "until someone notices" to "until the next sync cycle".

1.5 Benefits

Benefit	Detail
Universal support	Every major OTA supports iCal. No partnership, contract or API key required.
Zero integration cost	The operator connects a channel in under a minute by pasting a URL.
No approval process	Direct OTA APIs require partner onboarding, certification and sometimes revenue minimums. iCal requires nothing.
Works with long-tail channels	Small regional OTAs and personal Google Calendars are covered by the same mechanism.
Low maintenance	The format is frozen. Feeds rarely break after they are working.

1.6 Limitations

These must be communicated inside the UI, not buried in documentation. Operators who believe sync is instant will be surprised by their first double booking.

Limitation	Impact
Polling, not push	Availability propagates on a delay. Both sides poll. Real-world latency is 15 minutes to 3 hours in each direction.
No guaranteed inbound frequency	We control how often we pull. We do not control how often Airbnb or Booking.com pulls our export. Most OTAs refresh every 1 to 24 hours and ignore cache headers.
Dates only, no times	Most OTA feeds publish all-day <code>VALUE=DATE</code> events. This is fine for overnight stays and unreliable for time-slotted tours where two departures share a date.
No pricing or inventory count	iCal carries availability only. It cannot express rates, minimum stay, seat count or per-slot capacity. A tour with 12 seats is either fully open or fully blocked.
No guest data	Feeds are anonymized. Names, contact details and payment status are stripped by most OTAs.
No acknowledgement	There is no confirmation that a channel actually read our feed. Failure is silent on the outbound side.
One-way per feed	Each URL moves data in one direction. Two-way sync requires two configured links per channel.

1.7 When APIs are better than iCal

Situation	Recommended approach
Time-slotted tours with multiple daily departures	Direct API or channel manager. iCal cannot represent a 09:00 departure being full while 14:00 is open.
Seat-level or capacity-based inventory	Direct API. iCal is binary per date.
Rate and pricing parity across channels	Direct API or channel manager. Out of scope for iCal entirely.
High-volume, low-margin inventory where a double booking is costly	Direct API for the top two channels, iCal for the tail.
Guest details needed for fulfilment (pickup, dietary, manifest)	Direct API. iCal never carries this.
A channel already offers a certified partner API and we have volume with them	Direct API.

Recommended positioning: iCal is the default for every channel and the permanent solution for the long tail. Direct APIs are added later for the two or three channels that drive the most volume. The two mechanisms must be able to coexist on the same listing, which is why the data model records a `source_type` on every blocked date.

2. User Stories

2.1 Export

ID	Story	Acceptance criteria
US-01	As a travel operator, I want to copy my listing's iCal URL so I can paste it into an OTA.	One-click copy. Toast confirmation. URL visible in full and selectable.
US-02	As a travel operator, I want to know when my feed was last fetched so I can tell whether a channel is actually reading it.	"Last fetched" timestamp per requesting user agent, or a generic last-fetched time if the agent is unknown.
US-03	As a travel operator, I want to regenerate my export URL if I think it has leaked.	Confirmation dialog explaining that every connected channel must be updated with the new URL.
US-04	As a travel operator, I want my export feed to include manual blocks, not just bookings, so my personal downtime is respected on every channel.	Manual blocks appear as <code>VEVENT</code> entries with a distinct summary.

2.2 Import

ID	Story	Acceptance criteria
US-05	As a travel operator, I want to connect my Airbnb calendar by pasting a URL.	Platform dropdown, URL field, live validation, save.
US-06	As a travel operator, I want to connect multiple platforms to the same listing.	Up to the configured maximum (default 10) per listing. Duplicate URLs rejected.
US-07	As a travel operator, I want my bookings synchronized automatically without opening the page.	Scheduler runs at the per-connection interval. Default 1 hour.
US-08	As a travel operator, I want to know immediately if syncing failed.	Error badge in the table, error banner on the page, email or in-app notification after the retry budget is exhausted.
US-09	As a travel operator, I want to sync a calendar right now.	"Sync Now" button, rate limited, with an inline progress state.
US-10	As a travel operator, I want to disconnect a calendar.	Confirmation dialog offering a choice to keep or release the dates that connection blocked.
US-11	As a travel operator, I want to see what each sync actually did.	Sync history with imported event count, blocked date count, errors and duration.
US-12	As a travel operator, I want to be warned when an imported event conflicts with a booking I already have.	Conflict notification listing the affected dates and both sources.

2.3 Team and administration

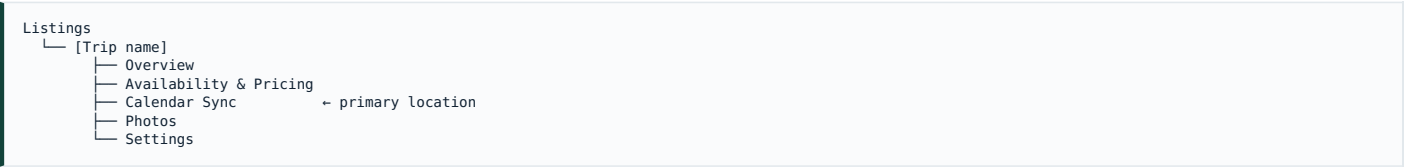
ID	Story	Acceptance criteria
US-13	As a property manager, I want to configure sync for listings I manage but not for listings I do not.	Permissions scoped by listing assignment.
US-14	As a staff member, I want to see sync status without being able to change it.	Read-only view. Action buttons hidden, not just disabled.
US-15	As a platform admin, I want to see failing connections across all accounts so I can spot a channel-wide outage.	Admin dashboard with failure counts grouped by platform.
US-16	As an account owner, I want an audit trail of who changed calendar configuration.	Audit log entries for add, edit, delete, regenerate and manual sync.

3. Information Architecture

3.1 Recommendation

Calendar Sync appears in **two places** with a shared underlying data model and two different scopes.

Primary (per listing):



Secondary (account-wide overview):



3.2 Why this split

An iCal connection binds one listing to one external calendar. It is not an account-level object. If the operator has 40 listings, an account-level-only design forces them to filter a 200-row table every time they touch a single trip. Putting the primary experience next to the listing keeps the mental model correct: *this trip's availability comes from these sources*.

The account-level view exists because operators with many listings need two things the per-listing view cannot give them:

- 1. A health overview. "Which of my 40 listings has a broken feed right now?"
- 2. Global defaults. Setting the sync interval and conflict policy 40 times is unacceptable.

The account-level page must be read-and-triage oriented. Every row links back to the per-listing page for editing. Only global defaults are edited at the account level.

3.3 Entry points

Entry point	Destination
Listing sidebar tab	Per-listing Calendar Sync
Availability calendar, "Blocked by Airbnb" chip	Per-listing Calendar Sync, scrolled to that connection
Settings → Integrations	Account-level dashboard
Sync failure notification (email or in-app)	Per-listing Calendar Sync, error banner expanded
Onboarding checklist item "Prevent double bookings"	Per-listing Calendar Sync, Add Calendar modal open

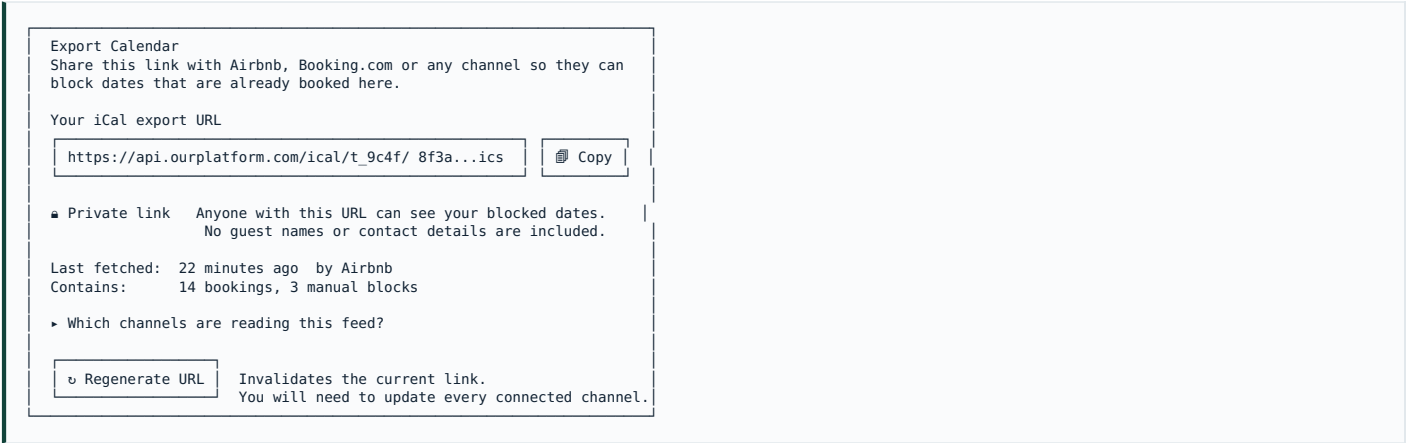
4. Screen Layout

4.1 Page structure (per listing)



The error banner (Section 0) renders only when at least one connection is in `error` or `invalid_url` state. It is dismissible per session but returns on reload while the condition persists.

4.2 Section A: Export Calendar



Element specification

Element	Behaviour
Export URL field	Read-only input, monospace, full width, text selectable, truncated with ellipsis in the middle on narrow viewports. Never wraps.
Copy button	Copies to clipboard, swaps icon to a checkmark for 2 seconds, fires a success toast. Falls back to <code>document.execCommand</code> on unsupported browsers.
Privacy note	Static, always visible. States plainly that the link is unguessable but public if leaked, and that no PII is included.
Last fetched	Relative time plus absolute time on hover. Sourced from access logs on the export endpoint. Shows "Never fetched" if there are no hits, with help text explaining that channels can take up to 24 hours for the first pull.
Contains	Live counts of bookings and manual blocks currently in the feed. Reassures the operator the feed is not empty.
Fetch log disclosure	Collapsed accordion. Expands to a small table of the last 20 fetches with timestamp, detected channel (from user agent) and response status.
Regenerate URL	Destructive-secondary styling. Opens a confirmation dialog.

Regenerate confirmation dialog

Regenerate export URL?

The current link will stop working immediately.
Any channel still using it will stop receiving updates and may resell dates you have booked.

Channels currently using this link:

- Airbnb fetched 22 min ago
- Booking.com fetched 4 hours ago

You will need to paste the new link into each one.

[Cancel] [Regenerate URL]

Grace period (implementation note): on regeneration, keep the old token valid for a configurable 24 hours in a `revoked_pending` state. Requests on the old token return the feed but are logged with a deprecation flag so the UI can show "Airbnb is still using your old link". This prevents an operator from silently breaking every channel by clicking one button. After the grace window the old token returns 410 Gone.

4.3 Section B: Imported Calendars

Imported Calendars [+ Add Calendar]
Dates booked on these channels will be blocked here automatically.

Platform	iCal URL	Last Sync	Status	Frequency	Actions
ⓐ Airbnb	.../calendar.ics	12 min ago 6 events	●Active	1 hour ▾	...
ⓑ Booking.com	.../ical?t=x9f	2 days ago 404 Not Found	●Error	1 hour ▾	...
ⓒ Viator	.../feed/882.ics	now	●Syncing	3 hours ▾	...
ⓓ GetYourGuide	.../ical/44a.ics	1 hour ago 0 events	○Paused	Daily ▾	...

4 of 10 calendars connected

Column specification

Column	Content	Notes
Platform	Channel logo (16px) plus name. "Other" shows a generic calendar icon plus the custom label.	Sortable.
iCal URL	Middle-truncated, monospace, with a copy affordance on hover. Full URL in a tooltip.	Never wraps the row.
Last Sync	Relative time. Second line shows the result: event count on success, error summary on failure.	Sortable. Absolute timestamp on hover.
Status	Badge. See table below.	Filterable.
Auto Sync Frequency	Inline dropdown, saves on change with an optimistic update and a toast.	Disabled when status is <code>paused</code> or <code>disconnected</code> .
Actions	Overflow menu (...) containing Sync Now, Edit, View Logs, Pause, Delete. Sync Now is also promoted as an icon button on hover for the primary path.	Delete is destructive and separated by a divider.

Status badges

Badge	Colour token	Meaning	Trigger
● Active	success	Last sync succeeded within the expected window	<code>last_success_at</code> within 2× the sync interval
● Syncing	info, animated	A sync job is running now	Job claimed, not yet finished
● Error	danger	Last sync failed and retries are exhausted	<code>consecutive_failures</code> >= <code>retry_limit</code>
● Invalid URL	danger	URL failed validation or returns non-calendar content	Validation failure at save time or on fetch
○ Paused	neutral	Auto sync disabled by the user	<code>auto_sync_enabled</code> = false
○ Disconnected	neutral	Connection soft-deleted, retained for history	<code>status</code> = <code>disconnected</code>
● Stale	warning	Last success is older than 2× the interval but not yet hard-failing	Computed at read time

Row states

- **Loading:** skeleton rows, four of them, matching final column widths.
- **Empty:** see below.
- **Error row:** left border in danger colour, error summary inline, "Fix" action promoted in place of "Sync Now".

Empty state

 #

No calendars connected yet

Connect the channels you sell on and we will block dates here automatically when a guest books somewhere else.

[+ Add your first calendar]

Not sure where to find your iCal link?
Airbnb · Booking.com · Expedia · Viator · GetYourGuide

Each channel name in the last line is a link that opens the relevant help drawer with screenshots of where that platform exposes its export URL. This is the single highest-leverage piece of UX in the whole feature. Most support tickets on iCal sync are "where do I get the link".

4.4 Add Calendar modal

Add Calendar

Platform *

Airbnb

shown only when Platform = "Other"

Platform name *

e.g. Local Tours Co

Calendar URL *

https://www.airbnb.com/calendar/ical/1234.ics?s=...

✓ Valid calendar feed. Found 6 events.

► Where do I find this on Airbnb?

Sync frequency

Every hour (recommended)

Notes (optional)

0 / 500

[Cancel]

[Test & Connect]

Behaviour

- Platform dropdown is searchable, with the six known channels pinned at the top, then alphabetical, then "Other".
- Selecting a known platform updates the URL placeholder and the contextual help link to that platform's instructions.
- Selecting "Other" reveals the free-text platform name field, which becomes required.
- URL validation runs client-side on blur (format, HTTPS, length) and server-side on a debounced 800ms pause (reachability, content type, parse check). Server-side validation is a dry run: it fetches and parses but writes nothing.
- The inline validation line has four states: neutral (empty), pending (spinner, "Checking feed..."), success (green, event count), failure (red, specific reason plus a fix hint).
- The primary button is labelled "Test & Connect" and is disabled until validation passes. It performs the first real sync immediately on success, then closes the modal with the row already in `syncing` state.
- If validation returns zero events, the button is enabled but a warning is shown: "This feed is valid but currently empty. That is normal if you have no upcoming bookings on this channel."

Edit mode reuses the same modal with the title "Edit Calendar", the platform locked (changing platform means deleting and re-adding, since attribution of existing blocked dates depends on it), and the primary button labelled "Save changes". If the URL changes, revalidation is mandatory before save.

4.5 Section C: Sync History

Sync History [All platforms ▼] [Last 7 days ▼]					
Date	Platform	Imported	Blocked	Errors	Duration
Aug 1, 14:02	Airbnb	6	+2 / -1	-	1.2s
Aug 1, 13:00	Airbnb	6	0	-	0.9s
Aug 1, 12:14	Booking	-	-	404 Not Found	4.0s
Aug 1, 12:00	Viator	3	+1	-	2.1s
Aug 1, 11:00	Booking	-	-	Timeout	30.0s

[Load more] Retained for 90 days

- **Blocked** shows the net change as added and removed counts. A sync that changes nothing shows 0, greyed. This is the column that tells the operator whether anything actually happened.
- **Errors** shows a short label. Clicking the row opens a detail drawer with the full error, the HTTP status, the response headers, the first 500 characters of the response body (sanitized) and the retry timeline.
- Rows with a conflict carry a warning icon next to the blocked count, linking to the conflict detail.
- Filters: platform, date range, outcome (all / success / failure / changed only).
- The account-level version of this table adds a Listing column.

4.6 Section D: Troubleshooting

Rendered as an accordion. Always present, never a separate page, because the operator who needs it is already looking at the error.

Where to find your iCal link One entry per supported channel, each with numbered steps and a screenshot. Airbnb, Booking.com, Expedia, Vrbo, Viator, GetYourGuide, MakeMyTrip, Google Calendar.

Supported formats

Accepted	Not accepted
https:// URLs returning text/calendar	http:// URLs
.ics files served with any content type, if the body parses	webcal:// (auto-rewritten to https:// with a notice)
Feeds with VEVENT blocks using DATE or DATE-TIME values	Google Calendar "secret address" in XML or HTML format
Feeds up to 5 MB	Feeds behind a login wall
UTF-8 and Latin-1 encoded feeds	ZIP or compressed calendar archives

Common errors and fixes

Error shown	What it means	Fix
Invalid URL format	The text is not a well-formed HTTPS URL	Re-copy the link. Make sure nothing was cut off.
404 Not Found	The channel deleted or rotated the link	Regenerate the export link on that channel and paste the new one here.
401 / 403	The feed requires a login	iCal feeds must be public. Use the export link, not the calendar page link.
Timeout	The channel did not respond in 30 seconds	Usually temporary. We retry automatically. If it persists over 24 hours, contact that channel.
Not a calendar feed	The URL returned a web page, not a .ics file	You probably copied the browser address bar. Look for a button labelled "Export calendar" or "Sync calendars".
Feed is empty	Valid, but contains no events	Normal if you have no bookings on that channel.
Malformed calendar data	The file is corrupted or truncated	Try again later. If it persists, report it to that channel.
Duplicate calendar	This exact URL is already connected to this listing	No action needed.
Feed too large	Over 5 MB	The channel is publishing years of history. Contact them or connect a narrower calendar.

FAQ

- How quickly do my dates update on Airbnb? We publish changes instantly, but each channel decides how often to read our feed. Most check every 1 to 24 hours.
- Will guests see my personal information? No. Feeds contain dates and a generic label only.
- Can I sync prices? Not with iCal. It carries availability only.
- What happens if two guests book the same date on two channels? We flag the conflict and notify you. The booking made on our platform is always kept. See Conflict policy in Settings.
- Do I need both import and export? Yes. Import stops others' bookings from being resold here. Export stops your bookings here from being resold elsewhere.

5. Complete User Flow

5.1 Happy path: connecting a calendar

1.	User opens Listings > Trip > Calendar Sync	
2.	Page loads. Export section renders with URL. Import table renders (empty state on first visit)	
3.	User clicks [+ Add Calendar]	
4.	Modal opens. Focus lands on Platform dropdown	
5.	User selects "Airbnb". Placeholder and help link update	
6.	User pastes the URL	
7.	Client validation passes (HTTPS, format, length)	
8.	After 800ms idle, server dry-run validation fires	
9.	Server fetches feed, checks size, content type, parses	
10.	Valid. Inline message: "Valid calendar feed. Found 6 events." [Test & Connect] enabled	
11.	User selects sync frequency (default: every hour)	
12.	User clicks [Test & Connect]	
13.	Connection persisted with status = syncing. Modal closes	
14.	Row appears in the table with an animated Syncing badge	
15.	First sync job runs immediately (not on the next scheduler tick)	
16.	Events parsed, deduplicated, written to blocked_dates	
17.	Availability calendar updated. Sync log written	
18.	Row flips to Active. Toast: "Airbnb connected. 6 dates blocked."	
19.	Connection enters the scheduler at the chosen interval	

5.2 Branch conditions

#	Branch	Trigger	System behaviour	User sees
B1	Invalid URL format	Fails regex or is not HTTPS	Reject before any network call	Inline red message, save blocked, fix hint
B2	webcal:// scheme	URL starts with webcal://	Auto-rewrite to https://, continue	Info note that the link was converted
B3	Unreachable host	DNS failure or connection refused	No connection created	"We could not reach this address. Check the link and try again."
B4	Network timeout	No response in 30s	No connection created on add. On scheduled sync, retry with backoff	On add, "The channel did not respond. Try again in a moment." On sync, Error badge after retries
B5	Authentication error	HTTP 401 or 403	No connection created	"This feed needs a login. iCal links must be public. Look for an Export or Sync option on that channel."
B6	Not found	HTTP 404 or 410	On add, reject. On an existing connection, mark error after retries	"This link no longer exists. Regenerate it on that channel."
B7	Server error	HTTP 5xx	Retry per backoff. Not a user fault	"That channel is having trouble. We will keep trying."
B8	Wrong content	Body is HTML, JSON or XML	Reject	"This is not a calendar feed. You may have copied the page address instead of the export link."
B9	Malformed ICS	Parser throws or no BEGIN:VCALENDAR	Reject on add. On an existing connection, error state, previous blocks retained	"We could not read this calendar file."
B10	Empty feed	Valid calendar, zero VEVENT	Allow. Create connection, status active, 0 events	Warning at add time, then "0 events" in the table, not an error
B11	Duplicate calendar	Normalized URL already exists on this listing and is not deleted	Block save	"This calendar is already connected." with a link to the existing row
B12	Same URL on a different listing	Normalized URL exists on another listing	Allow, but warn	"This link is also connected to [other trip]. Make sure that is intended."
B13	Max calendars reached	Count at listing limit	Disable Add Calendar	"You have reached the limit of 10 calendars for this listing."
B14	Feed too large	Content-Length or streamed size over 5 MB	Abort download	"This calendar is too large to sync (over 5 MB)."
B15	Too many events	Over 5,000 VEVENT after parse	Import the first 5,000 by start date ascending, flag partial	"Partial sync. This calendar has more events than we can process."
B16	Calendar removed at source	Previously working feed returns 404 for over 72 hours	Auto-pause, retain blocked dates, notify	Status Error, banner explaining dates are still blocked
B17	Manual sync	User clicks Sync Now	Enqueue at high priority, bypass interval, rate limit 1 per connection per 60s	Button becomes a spinner, badge goes Syncing, result toast
B18	Manual sync while a	Job already claimed	Reject the request	"A sync is already in progress."

	sync is running			
B19	Automatic sync	Scheduler tick, <code>next_sync_at <= now</code>	Claim job, run pipeline, reschedule	Nothing until the page is next loaded
B20	Deleted listing	Listing soft-deleted	Cancel scheduled jobs, set connections to disconnected, revoke export token after grace period	Connections disappear with the listing
B21	Conflict with an existing booking	Imported event overlaps a confirmed booking on our platform	Apply the conflict policy. Default is to keep our booking, record the conflict, do not block the date, notify	Warning icon in history plus a conflict notification
B22	Event removed upstream	An event present in the last sync is absent now	Release the blocked dates it created, unless a local booking now covers them	Blocked count shows a negative delta
B23	Disconnect with dates kept	User deletes and chooses "keep dates blocked"	Blocked dates reassigned to <code>source_type manual</code>	"Dates from Airbnb are now manual blocks."
B24	Disconnect with dates released	User deletes and chooses "release dates"	Blocked dates from that source deleted, availability recalculated	"Airbnb disconnected. 6 dates are now available."

5.3 Disconnect flow



6. Flowcharts

6.1 Export flow

MERMAID SOURCE

```
graph TD
    A[Booking created, modified or cancelled] --> B[Write to bookings table]
    B --> C[Emit availability.changed event]
    C --> D[Invalidate cached iCal feed for listing]
    D --> E[External platform requests feed]
    E --> F[GET /ical/:listing/:token.ics]
    F --> G[Token valid?]
    G --> H[404 Not Found]
    H --> I[Revoked, in grace period]
    I --> J[Serve feed and log deprecation flag]
    J --> K[Yes]
    K --> L[I[Cached feed present?]]
    L --> M[Yes]
    M --> N[J[Serve cached body]]
    N --> O[No]
    O --> P[K[Query bookings, manual blocks, imported blocks]]
    P --> Q[L[Build VCALENDAR, one VEVENT per occupied range]]
    Q --> R[M[Cache for 5 minutes]]
    R --> S[J]
    S --> T[H[Log fetch: timestamp, user agent, IP, status]]
    T --> U[N]
    U --> V[N]
    V --> W[O[Platform updates its own availability]]
```

What the export contains. The feed includes bookings made on our platform, manual blocks, and by default the blocks imported from other channels. Including imported blocks is what makes the hub topology work, since a date booked on Airbnb becomes unavailable on Booking.com by passing through us. Provide a per-connection setting to exclude one source from the export so a loop can be broken if a channel is found to be echoing our own feed back at us.

6.2 Import flow

MERMAID SOURCE

```
graph TD
    A[Scheduler tick, every 60 seconds] --> B[Select connections where next_sync_at is due]
    B --> C[Any due?]
    C --> D[No]
    D --> E[Z[Sleep until next tick]]
    E --> F[Yes]
    F --> G[D[Claim job with row lock, set status = syncing]]
    G --> H[HTTP GET feed, 30s timeout, 5MB cap, max 3 redirects]
    H --> I[Response OK?]
    I --> J[No]
    J --> K[ERR[Go to Error Flow]]
    K --> L[Yes]
    L --> M[G[Body parses as VCALENDAR?]]
    M --> N[No]
    N --> O[ERR]
    O --> P[Yes]
    P --> Q[H[Extract VEVENT list]]
    Q --> R[I[Normalize: expand RRULE, convert timezone, dedupe by UID]]
    R --> S[J[Compute checksum of normalized event set]]
    S --> T[K[Checksum equals last checksum?]]
    T --> U[Yes]
    U --> V[L[No-op sync, log zero changes]]
    V --> W[No]
    W --> X[M[Diff against stored calendar_events]]
    X --> Y[N[New events, insert]]
    Y --> Z[M[Changed events, update]]
    Z --> AA[M[Missing events, tombstone]]
    AA --> AB[N[Conflict detector]]
    AB --> AC[Q]
    AC --> AD[R[Overlaps a confirmed local booking?]]
    AD --> AE[Yes]
    AE --> AF[S[Apply conflict policy, record conflict, notify]]
    AF --> AG[No]
    AG --> AH[T[Upsert blocked_dates with source_connection_id]]
    AH --> AI[S]
    AI --> AJ[P[Release blocked dates for tombstoned events unless covered elsewhere]]
    AJ --> AK[T[Recalculate listing availability cache]]
    AK --> AL[U]
    AL --> AM[V[Write calendar_sync_logs row]]
    AM --> AN[L]
    AN --> AO[W]
    AO --> AP[X[Set status = active, reset failure counter]]
    AP --> AQ[Y[Schedule next_sync_at with jitter]]
    AQ --> AR[Z]
```

6.3 Error flow

MERMAID SOURCE

```
graph TD
    A[Sync attempt fails] --> B[Classify error]
    B --> C[Error class]
    C --> D[Transient: timeout, 5xx, DNS, reset]
    D --> E[D[Increment consecutive_failures]]
    E --> F[Permanent: 401, 403, 404, 410, malformed, wrong type]
    F --> G[E[Set status = error or invalid_url immediately]]
    G --> H[D[consecutive_failures below limit?]]
    H --> I[Yes]
    I --> J[G[Retry with exponential backoff and jitter: 1m, 5m, 15m, 1h, 4h]]
    J --> K[No]
    K --> L[E]
    L --> M[H[Status stays active, shown as Stale in UI]]
    M --> N[H[Retry attempt]]
    N --> O[I[Success?]]
    O --> P[Yes]
    P --> Q[K[Reset counter, status = active]]
    Q --> R[No]
    R --> S[A]
    S --> T[L[Write last error and sync log entry]]
    T --> U[M[Notification already sent for this error in 24h?]]
    U --> V[Yes]
    V --> W[N[Suppress, increment occurrence count]]
    W --> X[No]
    X --> Y[O[Send in-app and email notification]]
    Y --> Z[P[Show error banner on Calendar Sync page]]
    Z --> AA[N]
    AA --> AB[P]
    AB --> AC[Q[Failing for more than 72 hours?]]
    AC --> AD[Yes]
    AD --> AE[R[Auto-pause, retain blocked dates, send escalation notice]]
    AE --> AF[No]
    AF --> AG[S[Continue slow retries every 6 hours]]
```

Availability is retained on failure, deliberately. If a feed breaks, dates it previously blocked are not released. Releasing them would expose the operator to double bookings at exactly the moment they have least visibility. Dates stay blocked until the operator disconnects and explicitly chooses to release them.

6.4 Sync lifecycle

```
MERMAID SOURCE

stateDiagram-v2
    [*] --> Draft: User opens Add Calendar
    Draft --> Validating: URL entered
    Validating --> Draft: Validation failed
    Validating --> Validated: Feed reachable and parseable
    Validated --> Syncing: Test and Connect clicked
    Syncing --> Active: First sync succeeded
    Syncing --> Error: First sync failed
    Active --> Syncing: Scheduled tick or manual sync
    Active --> Paused: User pauses auto sync
    Active --> Stale: Missed two expected windows
    Stale --> Active: Sync succeeds
    Stale --> Error: Retry budget exhausted
    Error --> Syncing: Manual retry or backoff retry
    Error --> Paused: Auto-paused after 72h of failure
    Paused --> Syncing: User resumes or syncs manually
    Active --> Disconnected: User deletes connection
    Error --> Disconnected: User deletes connection
    Paused --> Disconnected: User deletes connection
    Disconnected --> [*]: Purged after 90 days
```

7. UI Components

7.1 Component inventory

Category	Component	Where used	Notes
Layout	Page header with breadcrumb	Top of page	Contains the Sync All action
Layout	Section card	Export, Import, History, Troubleshooting	Title, description, optional header action
Layout	Responsive grid	Desktop only, collapses below 1024px	
Data	Data table	Import list, sync history, fetch log	Sortable, filterable, paginated
Data	Definition list	Export metadata	
Data	Accordion	Troubleshooting, fetch log, help drawers	
Input	Read-only copy field	Export URL	Monospace, middle truncation
Input	Text input	Calendar URL, other platform name	With an inline validation slot
Input	Textarea with counter	Notes	500 character limit
Input	Select or combobox	Platform, frequency, filters	Searchable above 8 options
Input	Inline select	Frequency inside a table row	Saves on change
Input	Radio group	Disconnect dialog, conflict policy	
Input	Toggle	Auto sync, email notifications	
Action	Primary button	Test & Connect, Save changes	
Action	Secondary button	Cancel, Add Calendar	
Action	Destructive button	Regenerate URL, Disconnect	
Action	Icon button	Copy, Sync Now, overflow	Requires an accessible label
Action	Overflow menu	Row actions	Keyboard navigable
Action	Link button	Help links, View details	
Feedback	Status badge	Import table	7 variants
Feedback	Success toast	Copy, connect, sync complete, settings saved	Auto-dismiss after 4s
Feedback	Error toast	Failed action	Manual dismiss
Feedback	Inline validation message	URL field	4 states
Feedback	Alert banner	Page-level errors, partial sync warning	Dismissible per session
Feedback	Progress indicator	Sync All, first sync	Determinate when the count is known
Feedback	Spinner	Inline validation, Sync Now	
Feedback	Skeleton loader	Table rows, export card	Matches final dimensions
Feedback	Empty state	No calendars, no history, no filter results	Distinct copy per case
Overlay	Modal	Add and Edit Calendar	Focus trapped, ESC closes
Overlay	Confirmation dialog	Delete, Regenerate	Destructive action on the right
Overlay	Side drawer	Sync log detail, platform help	
Overlay	Tooltip	Truncated URLs, absolute timestamps, disabled reasons	
Overlay	Popover	Channels reading this feed	
Nav	Tabs	Listing sub-navigation	
Nav	Filter bar	History filters	

7.2 Loading and skeleton behaviour

Trigger	Treatment
Initial page load	Skeleton for the export card and 4 table rows. History loads lazily on scroll into view.
Inline URL validation	Spinner in the validation slot, button disabled, field stays editable
Sync Now	Row badge becomes Syncing, the action becomes a spinner, other rows stay interactive
Sync All	Header button shows progress such as "Syncing 2 of 4", all affected rows show Syncing

Frequency change	Optimistic update, silent revert plus error toast on failure
History pagination	Skeleton rows appended below the existing content

7.3 Accessibility requirements

- Status badges carry text, never colour alone. Screen reader text includes the reason, for example "Error, feed returned 404".
- The import list is a real table with proper header scopes, not a div grid.
- The copy button announces "Copied to clipboard" through an `aria-live="polite"` region.
- Status changes announce politely. Failures announce assertively.
- Modal focus is trapped and returns to the trigger on close.
- Every icon-only button carries an `aria-label`.
- Contrast meets WCAG 2.2 AA, 4.5:1 for text and 3:1 for interface boundaries.
- Every action is reachable by keyboard, including the row overflow menu and the inline frequency select.

7.4 Responsive behaviour

Breakpoint	Layout
1280px and above	Full table, all six columns
1024 to 1279px	URL column truncates harder, frequency uses short labels such as "1h"
768 to 1023px	Table collapses to stacked cards. Platform and status in the header, URL and last sync in the body, actions in a footer row
Below 768px	Single column cards. The export URL wraps to two lines with the copy button beneath. The modal becomes a full-screen sheet.

8. Field Specifications

8.1 Export section

Field	Type	Placeholder	Validation	Required	Example	Help text	Limits
Export URL	Read-only text	n/a	System generated	n/a	<code>https://api.ourplatform.com/ical/t_9c4f/8f3a1d7e2b6c4a90.ics</code>	Share this with any channel so it can block your booked dates.	512 chars
Export token	System, hidden	n/a	32-character URL-safe random from a cryptographic source	Yes	<code>8f3a1d7e2b6c4a90...</code>	Not displayed separately.	Exactly 32 chars
Last fetched	Read-only display	n/a	n/a	n/a	<code>22 minutes ago by Airbnb</code>	Channels decide how often they read your feed.	n/a
Include imported blocks	Toggle	n/a	Boolean	No	On	When on, dates blocked by other channels also appear in this feed.	n/a

8.2 Add and Edit Calendar modal

Field	Type	Placeholder	Validation	Required	Example	Help text	Limits
Platform	Select	Choose a platform	Must be in the enum	Yes	Airbnb	Pick the channel this calendar comes from.	airbnb, booking_com, expedia, vrbo, viator, getyourguide, makemytrip, google_calendar, other
Other platform name	Text	e.g. Local Tours Co	Required only when platform is other . Letters, digits, spaces and . - & ' . No URLs. Trimmed.	Conditional	Local Tours Co	This is only a label. It does not change how syncing works.	2 to 60 chars
Calendar URL	URL	<code>https://www.airbnb.com/calendar/ical/12345.ics?s=...</code>	Well-formed absolute URL. Scheme must be https , with webcal auto-rewritten. Host must be public, so no localhost, no private ranges, no .local . No credentials embedded. Must pass a live fetch and parse. Unique per listing after normalization.	Yes	<code>https://www.airbnb.com/calendar/ical/12345.ics?s=abc123</code>	Paste the export or sync link from that channel, not the address of the calendar page.	12 to 2048 chars
Sync frequency	Select	Every hour (recommended)	Must be in the allowed enum. Sub-hourly options gated by plan.	Yes	1 hour	More frequent syncing shortens the double-booking window.	15m, 30m, 1h, 3h, 6h, 12h, 24h
Notes	Textarea	Anything you want to remember about this connection	Plain text, HTML stripped	No	Main Airbnb listing, EU account	Only visible to your team.	0 to 500 chars
Auto sync enabled	Toggle	n/a	Boolean	No	On	Turn off to keep the connection but stop scheduled syncing.	n/a

8.3 Settings fields

Field	Type	Default	Validation	Required	Help text
Default sync frequency	Select	1 hour	Enum as above	Yes	Applied to new connections. Existing connections keep their own setting.
Timezone	Searchable select	Listing timezone	Valid IANA identifier	Yes	Used to interpret all-day events from channels that do not specify a timezone.
Retry attempts	Number stepper	5	Integer from 1 to 10	Yes	How many times to retry before marking a calendar as failed.
Conflict policy	Radio group	Keep our booking	Enum	Yes	What to do when an imported date is already booked here.
Email notifications	Toggle group	Failures on, successes off	Boolean per event type	No	Choose which sync events send an email.
Notification recipients	Multi-select of team members	Account owner	Users holding calendar.view	Yes	Who receives sync alerts.
Failure alert threshold	Select	After 3 failed attempts	immediately, 3, 5	Yes	How quickly to alert you when a feed breaks.

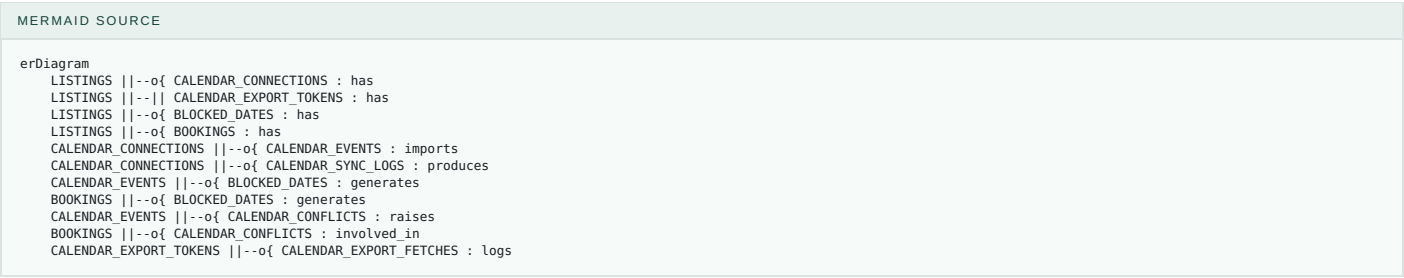
8.4 URL normalization rules

Applied before duplicate checking and before storage. Store the raw URL for display and fetching, and the normalized form for comparison.

1. Trim whitespace, strip zero-width and control characters.
2. Rewrite webcal:// to https:// .
3. Lowercase the scheme and host. Preserve path and query case, since OTA tokens are case sensitive.
4. Remove the default port :443 .
5. Remove a trailing slash only when the path is empty.
6. Preserve every query parameter and its order. Never strip anything that could be a token.
7. Remove the fragment.
8. Percent-encode consistently using uppercase hex.

9. Database Design

9.1 Entity relationship overview



9.2 calendar_connections

One row per external calendar attached to a listing.

Column	Type	Constraints	Notes
id	UUID	PK	
account_id	UUID	FK accounts, NOT NULL, indexed	Tenant scope, denormalized for query speed
listing_id	UUID	FK listings, NOT NULL, indexed	
platform	ENUM	NOT NULL	airbnb, booking_com, expedia, vrbo, viator, getyourguide, makemytrip, google_calendar, other
platform_label	VARCHAR(60)	NULL	Required when platform is other
ical_url	TEXT	NOT NULL	Raw URL as entered, encrypted at rest
ical_url_normalized	TEXT	NOT NULL	Used for duplicate detection
url_hash	CHAR(64)	NOT NULL, indexed	SHA-256 of the normalized URL, used in the unique index
status	ENUM	NOT NULL, default pending	pending, syncing, active, error, invalid_url, paused, disconnected
auto_sync_enabled	BOOLEAN	NOT NULL, default true	
sync_interval_minutes	SMALLINT	NOT NULL, default 60	15, 30, 60, 180, 360, 720, 1440
next_sync_at	TIMESTAMP TZ	NULL, indexed	Null when paused or disconnected
last_synced_at	TIMESTAMP TZ	NULL	Last attempt, successful or not
last_success_at	TIMESTAMP TZ	NULL	Drives the Stale computation
last_error_code	VARCHAR(50)	NULL	Machine readable, for example HTTP_404
last_error_message	TEXT	NULL	Human readable, shown in the UI
consecutive_failures	SMALLINT	NOT NULL, default 0	
last_checksum	CHAR(64)	NULL	Skips diffing when the feed is unchanged
last_event_count	INTEGER	NOT NULL, default 0	
exclude_from_export	BOOLEAN	NOT NULL, default false	Loop-breaking control
notes	VARCHAR(500)	NULL	
created_by	UUID	FK users, NOT NULL	
created_at	TIMESTAMP TZ	NOT NULL	
updated_at	TIMESTAMP TZ	NOT NULL	
deleted_at	TIMESTAMP TZ	NULL	Soft delete

Indexes

```
UNIQUE (listing_id, url hash) WHERE deleted_at IS NULL
INDEX (next_sync_at) WHERE auto_sync_enabled = true AND deleted_at IS NULL
INDEX (account_id, status)
INDEX (listing_id) WHERE deleted_at IS NULL
```

9.3 calendar_export_tokens

Separated from listings so that rotation keeps history and the grace period is representable.

--	--	--	--

Column	Type	Constraints	Notes
id	UUID	PK	
listing_id	UUID	FK listings, NOT NULL, indexed	
token	CHAR(32)	NOT NULL, UNIQUE	URL-safe random
token_hash	CHAR(64)	NOT NULL, UNIQUE, indexed	Lookup is done on the hash
status	ENUM	NOT NULL	active, revoked_pending, revoked
include_imported_blocks	BOOLEAN	NOT NULL, default true	
revoked_at	TIMESTAMPTZ	NULL	
expires_at	TIMESTAMPTZ	NULL	Set on revoke, now plus 24 hours
created_by	UUID	FK users	
created_at	TIMESTAMPTZ	NOT NULL	

Constraint: at most one row per listing with status `active` .

9.4 calendar_events

The parsed state of the last successful sync per connection. Acts as the diff baseline.

Column	Type	Constraints	Notes
id	UUID	PK	
connection_id	UUID	FK calendar_connections, NOT NULL, indexed, ON DELETE CASCADE	
listing_id	UUID	FK listings, NOT NULL, indexed	Denormalized
external_uid	VARCHAR(512)	NOT NULL	The <code>UID</code> property from the feed
summary	VARCHAR(255)	NULL	Usually "Reserved", "Not available", "CLOSED"
description_hash	CHAR(64)	NULL	Change detection without storing free text
start_date	DATE	NOT NULL, indexed	
end_date	DATE	NOT NULL	Exclusive, matching iCal semantics
start_time	TIME	NULL	Populated for <code>DATE-TIME</code> events
end_time	TIME	NULL	
is_all_day	BOOLEAN	NOT NULL, default true	
source_timezone	VARCHAR(64)	NULL	From <code>TZID</code> or the calendar-level <code>X-WR-TIMEZONE</code>
recurrence_rule	TEXT	NULL	Raw <code>RRULE</code> , retained for audit
is_recurrence_instance	BOOLEAN	NOT NULL, default false	Expanded occurrences
parent_event_id	UUID	FK calendar_events, NULL	Links an expansion to its master
raw_event	TEXT	NULL	Original <code>VEVENT</code> , kept for debugging, truncated at 4 KB
first_seen_at	TIMESTAMPTZ	NOT NULL	
last_seen_at	TIMESTAMPTZ	NOT NULL, indexed	Events not seen in the latest sync are tombstoned
removed_at	TIMESTAMPTZ	NULL	Tombstone marker

Indexes

```
UNIQUE (connection_id, external_uid, start_date) WHERE removed_at IS NULL
INDEX (listing_id, start_date, end_date)
INDEX (connection_id, last_seen_at)
```

The unique key includes `start_date` because some channels reuse a `UID` across recurring or amended reservations.

9.5 blocked_dates

The single availability layer read by search, the booking engine and the export builder. Every source writes here.

Column	Type	Constraints	Notes
id	UUID	PK	
listing_id	UUID	FK listings, NOT NULL, indexed	

blocked_date	DATE	NOT NULL, indexed	One row per calendar day, which keeps overlap queries trivial
source_type	ENUM	NOT NULL	booking, ical_import, manual, api_channel, maintenance
source_connection_id	UUID	FK calendar_connections, NULL	Set when source_type is ical_import
source_event_id	UUID	FK calendar_events, NULL	
source_booking_id	UUID	FK bookings, NULL	
slot_id	UUID	FK trip_slots, NULL	For time-slotted tours. Null blocks the whole day
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Indexes

```

UNIQUE (listing_id, blocked_date, source_type, COALESCE(source_connection_id, '00000000-...'), COALESCE(slot_id, '00000000-...'))
INDEX (listing_id, blocked_date)
INDEX (source_connection_id)

```

Design note. Storing one row per day rather than a range makes availability lookups a simple date-range scan and lets several sources block the same day independently. Releasing one source's block never accidentally opens a day another source still holds. For very long horizons this is more rows, but a 3-year horizon per listing is roughly 1,100 rows in the worst case, which is trivial.

9.6 calendar_sync_logs

Column	Type	Constraints	Notes
id	UUID	PK	
connection_id	UUID	FK calendar_connections, NOT NULL, indexed	
listing_id	UUID	FK listings, NOT NULL, indexed	
trigger	ENUM	NOT NULL	scheduled, manual, on_connect, retry, admin
triggered_by	UUID	FK users, NULL	Null for scheduled runs
status	ENUM	NOT NULL	success, partial, failed, skipped_unchanged
http_status	SMALLINT	NULL	
events_found	INTEGER	NOT NULL, default 0	
events_added	INTEGER	NOT NULL, default 0	
events_updated	INTEGER	NOT NULL, default 0	
events_removed	INTEGER	NOT NULL, default 0	
dates_blocked	INTEGER	NOT NULL, default 0	
dates_released	INTEGER	NOT NULL, default 0	
conflicts_detected	INTEGER	NOT NULL, default 0	
error_code	VARCHAR(50)	NULL	
error_message	TEXT	NULL	
response_size_bytes	INTEGER	NULL	
duration_ms	INTEGER	NOT NULL	
started_at	TIMESTAMPTZ	NOT NULL, indexed	
finished_at	TIMESTAMPTZ	NOT NULL	

Partition by month. Retain 90 days on standard plans, 12 months on higher tiers.

9.7 calendar_conflicts

Column	Type	Constraints	Notes
id	UUID	PK	
listing_id	UUID	FK listings, NOT NULL, indexed	
connection_id	UUID	FK calendar_connections, NOT NULL	The importing side
event_id	UUID	FK calendar_events, NULL	
booking_id	UUID	FK bookings, NULL	The local side

conflict_start	DATE	NOT NULL	
conflict_end	DATE	NOT NULL	
resolution	ENUM	NOT NULL	kept_local, blocked_anyway, manual_review
resolved_at	TIMESTAMPZ	NULL	
resolved_by	UUID	FK users, NULL	
acknowledged_at	TIMESTAMPZ	NULL	Clears the UI warning
created_at	TIMESTAMPZ	NOT NULL	

9.8 calendar_export_fetches

Powers the "last fetched by" display and abuse detection.

Column	Type	Notes
id	BIGSERIAL	PK
token_id	UUID	FK calendar_export_tokens, indexed
listing_id	UUID	Denormalized
fetched_at	TIMESTAMPZ	Indexed
user_agent	VARCHAR(255)	Truncated
detected_platform	VARCHAR(50)	Best-effort match from the user agent
ip_hash	CHAR(64)	Hashed, never stored raw
http_status	SMALLINT	
used_revoked_token	BOOLEAN	Drives the "still using the old link" warning

High write volume. Roll up daily into an aggregate table and retain raw rows for 30 days.

9.9 calendar_settings

Account-level and listing-level defaults. Listing values override account values. Nulls fall through to the account row.

Column	Type	Notes
id	UUID	PK
account_id	UUID	FK accounts, NOT NULL
listing_id	UUID	FK listings, NULL. Null means the account default
default_sync_interval_minutes	SMALLINT	
timezone	VARCHAR(64)	IANA identifier
retry_attempts	SMALLINT	
conflict_policy	ENUM	keep_local, block_anyway, manual_review
notify_on_failure	BOOLEAN	
notify_on_success	BOOLEAN	
notify_on_conflict	BOOLEAN	
failure_alert_threshold	SMALLINT	
notification_recipients	UUID[]	

10. Backend Architecture

10.1 Component map



10.2 Export endpoint

Public, unauthenticated in the session sense, authorized by an opaque token in the path. Stateless and cacheable.

- Resolves the token by hash. Rejects unknown or fully revoked tokens with 404, never 401, so the endpoint reveals nothing about which listings exist.
- Reads from the feed cache, keyed by listing plus token settings. Cache TTL of 5 minutes, invalidated eagerly on any availability change.
- Builds one `VEVENT` per contiguous occupied range, not per day, so that feeds stay compact.
- Emits `DTSTAMP`, `UID`, `DTSTART;VALUE=DATE`, `DTEND;VALUE=DATE`, `SUMMARY`. No guest data, no prices, no descriptions containing PII.
- `UID` is deterministic and stable across rebuilds, derived from the source type and source id, so consuming platforms treat it as the same event across fetches.
- Returns `Content-Type: text/calendar; charset=utf-8` and a `Content-Disposition` filename.
- Honours conditional requests using `ETag` and `Last-Modified`, returning 304 where possible. Many channels ignore this, so it is an optimization rather than a guarantee.
- Emits a `calendar_export_fetches` row asynchronously, off the response path.
- Horizon: past 30 days plus the next 24 months by default, configurable.

10.3 Import service

A pipeline of small, individually testable stages. Each stage takes the previous stage's output and never touches the database except through the persistence stages.

Stage	Responsibility	Failure behaviour
Fetcher	HTTP GET with a 30s total timeout, 10s connect timeout, maximum 3 redirects, 5 MB streamed cap, SSRF guard on every hop, custom user agent, <code>Accept: text/calendar</code>	Classify as transient or permanent, hand to the error path
Parser	Unfold lines per RFC 5545, tolerate CRLF and LF, tolerate missing <code>VERSION</code> , extract <code>VEVENT</code> blocks, detect and transcode non-UTF-8 encodings	Permanent failure <code>MALFORMED_ICS</code>
Normalizer	Resolve timezones, expand <code>RRULE</code> within the sync horizon, cap expansion at 200 instances per rule, convert <code>DATE-TIME</code> to the listing timezone, deduplicate by <code>UID</code> , drop events entirely in the past beyond the horizon, drop <code>STATUS:CANCELLED</code> events, drop zero-length events	Skip the individual event, count it, continue
Diff engine	Compare the normalized set against <code>calendar_events</code> , produce add, update and tombstone sets. Short circuit when the payload checksum is unchanged	n/a
Conflict detector	For every candidate date, check for a confirmed local booking or an existing block from another source, apply the policy	Records a conflict, never blocks the pipeline
Availability updater	Transactional upsert into <code>blocked_dates</code> , release tombstoned ranges, refresh the availability cache	Rolls back the whole sync, marks it failed
Logger	Write <code>calendar_sync_logs</code> , update the connection's denormalized state	Best effort, never fails the sync
Notifier	Evaluate notification rules, deduplicate, dispatch	Best effort, retried separately

10.4 Sync scheduler

- A single leader ticks every 60 seconds and selects connections where `next_sync_at <= now()`, `auto_sync_enabled = true`, `deleted_at IS NULL`, status in (active, error, pending).
- Jobs are claimed with `SELECT ... FOR UPDATE SKIP LOCKED` so several workers can drain the queue without collisions.
- Every scheduled time carries jitter of plus or minus 10 percent, which prevents thousands of connections created in one migration from stampeding the same OTA at the same second.

- Per-destination-host concurrency limits and a token bucket per host protect us from being rate limited by a channel. Airbnb feeds across all tenants share one bucket.
- Manual syncs go to a separate high-priority queue so a user-initiated action is never stuck behind a scheduled backlog.
- A stuck-job reaper resets any connection left in `syncing` for over 10 minutes.
- Idempotency: running the same sync twice produces the same end state, because the pipeline is a full-state diff rather than an incremental apply.

10.5 Conflict detector

An imported event conflicts when its date range overlaps a date already held by a confirmed local booking.

Policy	Behaviour	Suitable for
<code>keep_local</code> (default)	Do not create the block. Record a conflict. Notify. Our booking stands.	Almost all operators. Our platform is the source of truth.
<code>block_anyway</code>	Create the block alongside the booking. Record a conflict. Notify loudly.	Operators who want maximum caution and will resolve manually.
<code>manual_review</code>	Stage the block in a pending state, do not apply it, require an explicit decision in the UI.	Operators with high-value inventory.

Blocks from two different import sources covering the same date are not a conflict. They are independent rows and both are retained.

10.6 Retry strategy

Attempt	Delay after the previous attempt	Applies to
1	1 minute	Transient errors only
2	5 minutes	
3	15 minutes	
4	1 hour	
5	4 hours	
After 5	Move to <code>error</code> , then slow retry every 6 hours for 72 hours	
After 72 hours	Auto-pause, escalation notification, no further automatic attempts	

All delays carry jitter. Permanent errors (401, 403, 404, 410, malformed body, wrong content type) skip the ladder and move to `error` or `invalid_url` on the first occurrence, since retrying a 404 sixteen times helps nobody.

A circuit breaker operates per destination host. If more than 50 percent of requests to a host fail across all tenants within 5 minutes, the breaker opens for 10 minutes, connections are marked with a platform-outage flag, and the UI shows "Airbnb appears to be having an outage" rather than blaming the operator's URL.

10.7 Observability

- Metrics: sync duration histogram by platform, success ratio by platform, queue depth, feed size distribution, parse failure counts by reason, export fetch rate by platform, conflicts per day.
- Alerts: success ratio for any platform below 90 percent over 30 minutes, queue depth above threshold, export endpoint p99 latency, circuit breaker opened.
- Every sync carries a trace id that appears in the sync log detail drawer, so support can move from a user complaint to a trace in one step.

11. API Specification

All authenticated endpoints require a session or bearer token, are scoped to the caller's account, and enforce the permission matrix in section 13. All responses are JSON except the export feed.

11.1 GET /api/v1/trips/{trip_id}/calendar.ics

Not this one. The public feed is served from a separate unauthenticated route to keep it cacheable and free of session middleware.

11.2 GET /ical/{listing_slug}/{token}.ics

Purpose	Serve the public export feed for one listing
Auth	None. The token is the credential
Permissions	n/a
Rate limit	60 requests per hour per token, 600 per hour per IP
Request	No body. Supports If-None-Match and If-Modified-Since
Response 200	text/calendar; charset=utf-8, a full VCALENDAR document. Headers: ETag, Last-Modified, Cache-Control: public, max-age=300
Response 304	Not modified
Response 404	Unknown, revoked or expired token, or the listing is deleted. Body is empty
Response 410	Token was revoked and the grace period has ended
Response 429	Rate limited, with Retry-After

11.3 GET /api/v1/listings/{listing_id}/calendar-connections

Purpose	List every calendar connection on a listing
Auth	Required
Permissions	calendar.view
Query params	status, include_deleted (default false)
Response 200	Array of connection objects with id, platform, platform_label, ical_url_masked, status, sync_interval_minutes, last_synced_at, last_success_at, last_error_code, last_error_message, last_event_count, blocked_date_count, auto_sync_enabled, notes
Errors	401 unauthenticated, 403 no access to the listing, 404 listing not found

The URL is masked by default. The full URL is returned only to roles holding calendar.manage, since the token in the query string is a credential for the other platform.

11.4 POST /api/v1/calendar-connections/validate

Purpose	Dry-run validation for the Add Calendar modal. Writes nothing
Auth	Required
Permissions	calendar.manage
Rate limit	20 per minute per user
Request	{ "listing_id": "...", "ical_url": "https://..." }
Response 200	{ "valid": true, "event_count": 6, "calendar_name": "Airbnb (Bali Villa)", "first_event": "2026-08-14", "last_event": "2026-11-02", "feed_size_bytes": 4102, "warnings": ["EMPTY_FEED"]} }
Response 422	{ "valid": false, "error_code": "NOT_A_CALENDAR", "message": "...", "hint": "..." }
Errors	400 malformed body, 403, 409 duplicate, 429

11.5 POST /api/v1/calendar-connections

Purpose	Create a connection and trigger the first sync
Auth	Required
Permissions	calendar.manage

Request	<code>{ "listing_id", "platform", "platform_label", "ical_url", "sync_interval_minutes", "notes", "auto_sync_enabled" }</code>
Idempotency	Accepts an <code>Idempotency-Key</code> header
Response 201	The created connection object with <code>status: "syncing"</code>
Errors	400 validation failure with a field-level error map, 403, 409 <code>DUPLICATE_CALENDAR</code> , 422 <code>INVALID_FEED</code> , 429, 402 <code>PLAN_LIMIT_REACHED</code> when the listing is at its calendar cap

11.6 PATCH `/api/v1/calendar-connections/{id}`

Purpose	Update URL, interval, notes, auto-sync toggle or export exclusion
Auth	Required
Permissions	<code>calendar.manage</code>
Request	Partial object. <code>platform</code> is immutable
Behaviour	Changing <code>ical_url</code> forces revalidation and resets <code>last_checksum</code> , <code>consecutive_failures</code> and error state
Response 200	The updated connection
Errors	400, 403, 404, 409 duplicate, 422 invalid feed

11.7 DELETE `/api/v1/calendar-connections/{id}`

Purpose	Disconnect a calendar
Auth	Required
Permissions	<code>calendar.manage</code>
Query params	<code>release_dates</code> boolean, default false
Behaviour	Soft delete, cancel scheduled jobs, apply the date decision, write an audit entry
Response 200	<code>{ "id": "...", "dates_released": 6, "dates_converted_to_manual": 0 }</code>
Errors	403, 404, 409 if a sync is currently running

11.8 POST `/api/v1/calendar-connections/{id}/sync`

Purpose	Trigger an immediate sync
Auth	Required
Permissions	<code>calendar.sync</code>
Rate limit	1 per connection per 60 seconds, 10 per listing per hour
Request	Empty body
Response 202	<code>{ "job_id": "...", "status": "queued" }</code>
Response 409	<code>SYNC_IN_PROGRESS</code>
Errors	403, 404, 429 with <code>Retry-After</code>

Clients poll `GET /api/v1/calendar-sync-jobs/{job_id}` or subscribe to the websocket channel `listing:{id}:calendar` for a push update.

11.9 POST `/api/v1/listings/{listing_id}/calendar-connections/sync-all`

Purpose	Sync every active connection on a listing
Auth	Required
Permissions	<code>calendar.sync</code>
Rate limit	1 per listing per 5 minutes
Response 202	<code>{ "jobs": [{ "connection_id", "job_id" }] }</code>

11.10 GET `/api/v1/calendar-sync-logs`

Purpose	Sync history, per listing or per account
Auth	Required

Permissions	calendar.view
Query params	listing_id, connection_id, platform, status, from, to, cursor, limit (max 100)
Response 200	{ "data": [...], "next_cursor": "..." }
Errors	400 invalid range, 403

11.11 GET /api/v1/calendar-sync-logs/{id}

Returns the full log entry including the sanitized response excerpt, response headers, retry timeline and trace id. Requires calendar.manage , because the excerpt can contain the feed URL.

11.12 GET /api/v1/listings/{listing_id}/calendar-export

Purpose	Return the export URL and its metadata
Permissions	calendar.view for metadata, calendar.manage to see the full token URL
Response 200	{ "export_url", "token_created_at", "last_fetched_at", "last_fetched_by", "booking_count", "manual_block_count", "include_imported_blocks", "recent_fetches": [...] }

11.13 POST /api/v1/listings/{listing_id}/calendar-export/regenerate

Purpose	Rotate the export token
Permissions	calendar.manage
Request	{ "grace_period_hours": 24 }
Response 200	{ "export_url", "old_token_expires_at" }
Side effects	Audit log entry, notification to the account owner
Rate limit	5 per listing per day

11.14 GET /api/v1/calendar-conflicts and POST /api/v1/calendar-conflicts/{id}/acknowledge

List unresolved conflicts and mark one as reviewed. calendar.view to list, calendar.manage to acknowledge.

11.15 GET and PUT /api/v1/calendar-settings

Read and write account-level or listing-level defaults. PUT requires calendar.settings .

11.16 Error response shape

Uniform across every endpoint.

```
{
  "error": {
    "code": "INVALID_FEED",
    "message": "This is not a calendar feed.",
    "hint": "You may have copied the calendar page address instead of the export link.",
    "field": "ical_url",
    "trace_id": "01J9X..."
  }
}
```

The code drives UI logic. The message is displayed. The hint is shown as secondary text. Never surface raw parser exceptions or upstream response bodies to the client.

12. Validation Rules

12.1 URL validation

Rule	Level	Error code	Message
Not empty	Client	REQUIRED	Enter a calendar URL.
Length 12 to 2048	Client	URL_LENGTH	This link is too short to be valid. / This link is too long.
Parses as an absolute URL	Client	URL_MALFORMED	This does not look like a valid link.
Scheme is https after webcal rewrite	Client	URL_NOT_HTTPS	Calendar links must start with https.
No embedded credentials (user:pass@)	Client	URL_HAS_CREDENTIALS	Remove the username and password from this link.
Host is not an IP literal, localhost, .local , or a private or link-local range, checked again after every redirect	Server	URL_BLOCKED_HOST	This address cannot be used.
Host resolves	Server	HOST_UNREACHABLE	We could not reach this address.
Not already connected to this listing after normalization	Server	DUPLICATE_CALENDAR	This calendar is already connected.
Listing is below the calendar cap	Server	MAX_CALENDARS_REACHED	You have reached the limit of 10 calendars for this listing.

12.2 Feed validation

Rule	Threshold	Error code	Behaviour
Response status	2xx required	HTTP_{status}	401/403/404/410 permanent, 5xx transient
Redirects	Maximum 3, each re-checked against the SSRF guard	TOO_MANY_REDIRECTS	Permanent
Connect timeout	10 seconds	CONNECT_TIMEOUT	Transient
Total timeout	30 seconds	FEED_TIMEOUT	Transient
Feed size	5 MB, enforced on Content-Length and again while streaming	FEED_TOO_LARGE	Permanent, abort the download
Content type	text/calendar preferred. Other types accepted only if the body starts with BEGIN:VCALENDAR	NOT_A_CALENDAR	Permanent
Body starts with BEGIN:VCALENDAR after BOM and whitespace stripping	Required	MALFORMED_ICS	Permanent
Body ends with END:VCALENDAR	Required	TRUNCATED_FEED	Transient, likely a partial download
Event count	Maximum 5,000 after expansion	TOO_MANY_EVENTS	Partial import, warn
Recurrence expansion	Maximum 200 instances per rule, capped at the sync horizon	RECURRENCE_TRUNCATED	Warn only
Event count is zero	Allowed	EMPTY_FEED	Warning, not an error
Encoding	UTF-8, with Latin-1 fallback	ENCODING_ERROR	Permanent

12.3 Per-event validation

Events that fail these checks are skipped and counted. They never fail the whole sync.

Rule	Behaviour on failure
DTSTART present and parseable	Skip event
DTEND present, or derive from DTSTART plus DURATION , or default to a one-day event	Derive
DTEND after DTSTART	Skip event
Range within the sync horizon (past 30 days to plus 24 months)	Skip silently, do not count as an error
Range shorter than 730 days	Skip and warn, since a multi-year event usually signals a broken feed
UID present, or synthesize a deterministic one from a hash of start, end and summary	Synthesize
STATUS is not CANCELLED	Skip event
TRANSP is not TRANSPARENT	Skip event, since transparent events do not block time

12.4 Token validation

Rule	Behaviour
Token length exactly 32 and character set is URL-safe	Reject with 404 before any database lookup
Token hash exists	Otherwise 404
Token status is <code>active</code>	<code>revoked_pending</code> serves with a deprecation log entry. <code>revoked</code> returns 410
<code>expires_at</code> in the future or null	Otherwise 410
Parent listing exists and is not deleted	Otherwise 404

12.5 Settings validation

Field	Rule
Sync interval	Must be in the enum. Intervals below 60 minutes require the relevant plan tier, otherwise 402
Timezone	Must be a valid IANA identifier
Retry attempts	Integer 1 to 10
Notification recipients	Every user must belong to the account and hold <code>calendar.view</code>
Conflict policy	Must be in the enum

13. Permissions

13.1 Permission keys

Key	Grants
calendar.view	See the Calendar Sync page, connection statuses, masked URLs and sync history
calendar.sync	Trigger a manual sync or Sync All
calendar.manage	Add, edit and delete connections. See full URLs. Regenerate the export token
calendar.settings	Change account-level or listing-level calendar settings
calendar.audit	View the audit trail and full sync log detail
calendar.admin	Platform-wide visibility across accounts, force sync, override the circuit breaker

13.2 Role matrix

Capability	Owner	Manager	Staff	Read-only	Platform admin
View Calendar Sync page	Yes	Yes, assigned listings only	Yes, assigned listings only	Yes, assigned listings only	Yes, all accounts
Copy export URL	Yes	Yes	No, masked	No, masked	Yes
Regenerate export token	Yes	No	No	No	Yes, with a reason recorded
Add a calendar	Yes	Yes	No	No	Yes
Edit a calendar	Yes	Yes	No	No	Yes
Delete a calendar	Yes	Yes	No	No	Yes
See the full iCal URL	Yes	Yes	No	No	Redacted by default, revealable with an audit entry
Change sync frequency	Yes	Yes	No	No	Yes
Trigger a manual sync	Yes	Yes	Yes	No	Yes
View sync history	Yes	Yes	Yes	Yes	Yes
View full log detail	Yes	Yes	No	No	Yes
Change calendar settings	Yes	Listing level only	No	No	Yes
Acknowledge conflicts	Yes	Yes	Yes	No	Yes
Receive failure notifications	Yes	Yes	Optional	No	Via platform alerting
View the audit trail	Yes	No	No	No	Yes

13.3 Rules

- Listing scope is enforced first. A manager without an assignment to a listing gets 404, not 403, so listing existence is not leaked.
- Staff can trigger a manual sync but cannot change configuration. This matches how operations teams actually work, where front-desk staff need to force a refresh before confirming a walk-in but must not be able to detach a channel.
- Read-only roles see masked URLs. The masking is applied server-side. The full value never reaches a client that lacks the permission.
- Platform admin access to a customer's full iCal URL requires a justification field and writes an audit record visible to the account owner.
- Every mutation records actor, timestamp, IP, before value and after value in the audit trail.

14. Notifications

14.1 Notification matrix

Event	Channels	Default	Recipients	Priority	Deduplication
Sync successful (scheduled)	None	Off	n/a	n/a	Never notify. This fires hundreds of times a day
Manual sync complete	In-app toast	On	The initiating user only	Low	Immediate, no email
Sync failed, first failure	In-app	On	Users with <code>calendar.manage</code> on the listing	Low	Suppressed until the threshold is reached
Sync failed, threshold reached	In-app, email	On	Users with <code>calendar.manage</code>	High	One per connection per 24 hours
Calendar auto-paused after 72h	In-app, email	On, cannot be disabled	Account owner and managers	Critical	One per connection per pause
Invalid URL detected	In-app, email	On	Users with <code>calendar.manage</code>	High	One per connection per URL value
Duplicate booking or conflict detected	In-app, email, optional SMS	On, cannot be disabled	Account owner and managers	Critical	One per conflict, never batched away
Export token regenerated	In-app, email	On, cannot be disabled	Account owner	High	Immediate
Old export token still in use	In-app	On	Account owner	Medium	Once per day during the grace period
Export feed never fetched after 48h	In-app	On	Users with <code>calendar.manage</code>	Medium	Once per connection setup
Platform-wide outage detected	In-app banner	On	All affected accounts	Medium	One per outage window
Partial sync due to size or event caps	In-app	On	Users with <code>calendar.manage</code>	Medium	One per connection per 7 days

14.2 Content principles

Every failure notification answers three questions in this order: what broke, what it means for the business, what to do next.

Weak: "Sync failed for connection a4f9e2."

Correct:

Booking.com calendar stopped syncing Bali Sunrise Trek

We have not been able to read your Booking.com calendar since 30 July. The feed returns "not found", which usually means the link was regenerated on Booking.com.

The 6 dates it previously blocked are still blocked, so you are not at risk of a double booking right now. New Booking.com reservations will not appear here until this is fixed.

[Fix this calendar]

Rules:

- Name the listing and the platform in the subject line. An operator with 40 listings cannot act on a generic alert.
- Always state whether existing blocks are retained. This is the operator's first question and the answer is yes.
- Link directly to the affected connection, not the dashboard.
- Never blame the user for a channel-side outage.
- Batch across listings. If 12 connections fail because Airbnb is down, send one email listing all 12, not 12 emails.
- Respect quiet hours for non-critical notifications, using the account timezone. Conflicts and auto-pause events ignore quiet hours.

14.3 Digest

A weekly digest for accounts with more than 5 connections: syncs run, dates blocked, conflicts detected, connections needing attention, and any channel that has not fetched the export feed in over 7 days. Opt-out available.

15. Settings

15.1 Account level (Settings → Integrations → Calendar Sync)

Calendar Sync Settings

Default sync frequency

Every hour (recommended)

Applied to new calendars.

15 minutes

30 minutes

1 hour (recommended)

3 hours

6 hours

12 hours

Daily

Timezone

Asia/Dubai

Used for all-day events with no timezone.

Retry attempts before marking a calendar as failed

5

▲▼

When an imported date is already booked here

☒ Keep my booking and warn me

☐ Block the date anyway and warn me

☐ Hold it for my review before blocking

← recommended

Email me about

☒ Sync failures

☒ Double bookings and conflicts

☐ Successful syncs

☒ Weekly summary

(cannot be turned off)

Send alerts to

☒ Owner

☒ Ops team

▼

Quiet hours

☒ 22:00 to 07:00

Critical alerts ignore this.

[Save changes]

15.2 Setting reference

Setting	Scope	Default	Notes
Auto sync	Per connection	On	Turning it off keeps the connection and stops the schedule
Sync frequency	Per connection, defaulting from the account	1 hour	Sub-hourly is a plan-gated feature, both to control cost and because most channels do not update fast enough to justify it
Timezone	Account, overridable per listing	Listing timezone	Determines how all-day events map to calendar days
Retry attempts	Account	5	
Conflict policy	Account, overridable per listing	Keep our booking	
Email notifications	Account, per event type	Failures on	Conflict alerts cannot be disabled
Notification recipients	Account	Owner	
Quiet hours	Account	Off	Critical alerts override
Sync horizon	Account, admin-visible only	Past 30 days, next 24 months	Exposed to support, not to end users initially
Include imported blocks in export	Per listing	On	Loop-breaking control
Exclude a specific connection from export	Per connection	Off	Used when a channel echoes our own feed back
Maximum calendars per listing	Plan	10	

15.3 Frequency guidance shown inline

Interval	Best for	Trade-off
15 or 30 minutes	High-turnover inventory, several channels, short lead times	Higher risk of being rate limited by a channel. Most feeds do not change this often
1 hour	The default for almost everyone	Typical exposure window of under an hour on the inbound side
3 to 6 hours	Low-volume listings, long lead times	Meaningfully larger double-booking window

12 or 24 hours	Rarely booked or seasonal inventory	Only appropriate when bookings are days apart
----------------	-------------------------------------	---

The UI should state honestly that reducing our interval only shortens half the loop. How quickly Airbnb sees our changes is controlled by Airbnb.

16. Edge Cases

#	Case	Handling
E1	Listing deleted	Cancel scheduled jobs, set connections to <code>disconnected</code> , revoke the export token after a 24-hour grace period, retain sync logs for the retention window, hard-delete blocked dates with the listing
E2	Listing unpublished but not deleted	Syncing continues. Availability still matters for channels that keep the listing live
E3	Booking cancelled on our platform	Release its blocked dates, invalidate the feed cache, and let channels pick up the change on their next fetch. Log it, because an operator who cancels and immediately sees no change on Airbnb will open a ticket
E4	Imported event disappears upstream (guest cancelled on Airbnb)	Tombstone the event, release its dates unless another source or a local booking still holds them
E5	Recurring events (<code>RRULE</code>)	Expand within the horizon, cap at 200 instances per rule, store each instance with a parent link. <code>EXDATE</code> exclusions are honoured. Infinite rules are truncated at the horizon and flagged
E6	<code>RECURRENCE-ID</code> overrides	Treat as an exception instance that replaces the generated occurrence for that date
E7	Timezone mismatch	All-day events are taken as literal calendar dates with no conversion, which is what every OTA means by them. Timed events are converted from <code>TZID</code> (or <code>X-WR-TIMEZONE</code> , or the listing timezone as a last resort) into the listing timezone before mapping to days
E8	DST transition inside a range	Compute in the listing timezone using date arithmetic, never by adding fixed second counts
E9	Half-open interval confusion	iCal <code>DTEND</code> is exclusive. A stay of 14 to 18 August blocks the 14th, 15th, 16th and 17th. Some feeds get this wrong and publish an inclusive end. Detect single-day events where <code>DTSTART</code> equals <code>DTEND</code> and treat them as one blocked day rather than zero
E10	Duplicate events inside one feed	Deduplicate by <code>UID</code> . If <code>UID</code> is missing, deduplicate by a hash of start, end and summary
E11	Same physical booking arriving from two channels	Both create blocks. Not treated as a conflict. Both are retained and released independently
E12	Feedback loop, a channel republishing our own events	Detect our own deterministic <code>UID</code> prefix in an inbound feed and skip those events. Also offer the manual <code>exclude_from_export</code> control
E13	Platform downtime	Circuit breaker per host. The UI shows a platform outage message, not a user error. No notification spam during a known outage
E14	Slow response	30-second total timeout. Repeated slowness on the same host lowers the concurrency limit for that host rather than failing every connection
E15	Network failure mid-download	Treated as transient. Partial bodies are discarded, never parsed
E16	We are rate limited by a channel (429)	Honour <code>Retry-After</code> . Increase the effective interval for that host across all tenants. Never retry immediately
E17	Partial sync	Apply what parsed successfully, mark the log <code>partial</code> , warn the user, keep the previous state for the portion that failed
E18	Corrupted or truncated ICS	Reject the whole payload. Never apply a half-parsed calendar, since a truncated feed looks identical to a feed where all the later bookings were cancelled
E19	Very large calendars	5 MB and 5,000 event caps. Stream-parse rather than buffering the whole body. Import the earliest 5,000 events by start date, since near-term dates matter most
E20	Imported booking overlaps an existing booking	Apply the conflict policy. Default is to keep the local booking, record a conflict, do not block, and notify at critical priority
E21	Overlap with an existing manual block	Not a conflict. Both rows coexist
E22	Event entirely in the past	Ignored beyond the 30-day trailing window
E23	Event spanning the horizon boundary	Import the portion inside the horizon. Re-evaluate on each sync as the horizon rolls forward
E24	Feed served over a redirect chain to a different host	Allowed up to 3 hops, with the SSRF guard applied to every hop, and the final host used for rate limiting
E25	Feed with a self-signed or expired TLS certificate	Rejected. Presented as "This channel's security certificate is not valid" with no option to bypass
E26	Feed requiring a specific user agent	We send a stable identifying user agent with a contact URL. Some channels block generic clients. This is documented in troubleshooting
E27	Export token leaked publicly	Regenerate with a grace period. The fetch log helps identify unexpected consumers
E28	Time-slotted tour with per-slot inventory	iCal blocks the whole day, since the format cannot express slot-level availability. This must be stated in the UI when a listing has slots configured, with a recommendation to use a direct API for those channels
E29	Multi-day tour with shared departures	Same limitation as E28. Blocking is per date, not per departure
E30	Two team members editing the same connection	Optimistic concurrency on <code>updated_at</code> . The second save is rejected with 409 and a reload prompt
E31	Sync running while the user deletes the connection	Delete returns 409 while a job is in flight. The UI retries once the job settles, within a few seconds
E32	Clock skew between our servers and the channel	Never trust <code>DTSTAMP</code> for ordering. Diff on content, not timestamps

E33	Feed with no <code>VERSION</code> or <code>PRODID</code>	Tolerated. Only <code>BEGIN:VCALENDAR</code> and parseable <code>VEVENT</code> blocks are required
E34	Feed using <code>\n</code> instead of <code>\r\n</code>	Tolerated. The parser normalizes line endings before unfolding
E35	Non-UTF-8 encoding	Detect from the charset parameter or by sniffing. Fall back to Latin-1. Reject anything undecodable
E36	Account downgraded below its calendar limit	Existing connections keep working. New connections are blocked until the count is under the limit. Never silently disconnect a paying customer's channels

17. Security

17.1 Export token design

- 32 URL-safe characters from a cryptographically secure source, giving roughly 190 bits of entropy. Not guessable, not enumerable, not derived from the listing id.
- Stored as a SHA-256 hash. Lookups compare hashes. The raw token exists in the database only where it must be redisplayed to the owner, encrypted at rest with envelope encryption.
- Scoped to a single listing. One token never exposes another listing.
- Rotatable with a 24-hour grace window so that rotation does not silently break every channel.
- Never logged in plain text. Access logs and error traces mask the token segment of the path.
- Never included in `Referer`-leaking contexts. The export response sets `Referrer-Policy: no-referrer`.

17.2 Export endpoint hardening

Control	Implementation
Enumeration resistance	Invalid tokens return 404 with a constant-time comparison and a uniform response time
No PII	The feed contains dates, a stable <code>UID</code> and a generic summary. Never guest names, emails, phone numbers, prices or booking references
Rate limiting	60 requests per hour per token, 600 per hour per IP, with <code>Retry-After</code> on 429
Abuse detection	Alert when a single token is fetched by more than 20 distinct IPs in an hour, or when fetch volume jumps more than 10x week over week
Caching	Public cache headers are safe because the token is the credential. The CDN key includes the token
Headers	<code>X-Content-Type-Options: nosniff</code> , <code>Referrer-Policy: no-referrer</code> , no CORS headers, since browsers are not the intended consumer
Deleted listings	Immediate 404 regardless of token validity

17.3 Import security (this is where the real risk is)

Fetching a URL supplied by a user is a server-side request forgery primitive. Every control below is mandatory, not optional.

Control	Implementation
SSRF guard	Resolve the hostname, reject loopback, private (10/8, 172.16/12, 192.168/16), link-local (169.254/16, fe80::/10), unique-local (fc00::/7), multicast and reserved ranges. Re-check after every redirect. Use a resolved-IP-pinned connection to close the DNS rebinding window between check and connect
Metadata endpoints	Explicitly deny 169.254.169.254 and cloud provider metadata hostnames
Scheme restriction	Only <code>https</code> . No <code>file</code> , <code>gopher</code> , <code>ftp</code> , <code>data</code>
Redirect limit	Maximum 3 hops, each re-validated
Egress isolation	Fetches run from a dedicated worker pool with a network policy that permits outbound 443 only, and no access to internal services
Response size	Streamed 5 MB cap enforced before parsing
Parser hardening	No external entity resolution, no recursion on nested structures, bounded recurrence expansion, timeouts on the parse stage itself. Fuzz the parser in CI
Timeouts	10s connect, 30s total, enforced at the socket level
Content isolation	Feed content is treated as untrusted input. <code>SUMMARY</code> and <code>DESCRIPTION</code> are escaped on display and never rendered as HTML
URL storage	Encrypted at rest, since OTA iCal URLs contain bearer tokens for the operator's account on that platform
URL masking	Full URLs returned only to <code>calendar.manage</code> holders. Masked in logs, error messages, support tooling and analytics
Error leakage	Upstream response bodies are never returned verbatim. The log detail drawer shows a sanitized 500-character excerpt with anything token-shaped redacted

17.4 Rate limiting summary

Action	Limit
Export feed fetch	60 per hour per token, 600 per hour per IP
URL validation	20 per minute per user
Manual sync	1 per connection per 60 seconds, 10 per listing per hour

Sync all	1 per listing per 5 minutes
Token regeneration	5 per listing per day
Connection creation	30 per account per hour
Outbound fetch per destination host	Global token bucket shared across tenants

17.5 Audit trail

Every one of these writes an immutable audit record with actor, timestamp, IP, user agent, before value and after value:

connection created, connection updated (with a field-level diff), URL changed (values masked, only a change flag recorded), connection deleted (with the date-release decision), manual sync triggered, sync frequency changed, export token regenerated, calendar settings changed, conflict acknowledged or resolved, admin viewed a full iCal URL (with the stated justification).

Audit records are retained for 24 months, are visible to the account owner, and are exportable.

17.6 Data protection

- Guest personal data never enters an iCal feed in either direction. If an inbound feed contains a guest name in SUMMARY, we store the summary for debugging but never display it in the availability calendar and never re-export it.
- raw_event retention is capped at 4 KB per event and purged with the connection.
- IP addresses in fetch logs are hashed with a rotating salt.
- Under a deletion request, connections, events, logs and fetch records for the account are purged within the contractual window. Audit records are retained where legally required, with identifiers pseudonymized.

18. UX Recommendations

18.1 Principles

Make the invisible visible. Sync is a background process. The operator's trust depends entirely on being able to see that it ran, when it ran and what it changed. Every surface should answer "is this working right now" without a click.

Be honest about latency. The single largest source of support load on iCal features is an operator expecting instant propagation. Say plainly, in the UI, that channels read our feed on their own schedule and that changes typically appear within a few hours. Setting this expectation once at connection time prevents the ticket.

Fail loudly, degrade safely. A broken feed must be impossible to miss, and a broken feed must never silently open dates for sale.

18.2 Specific recommendations

Recommendation	Detail
One-click copy	The export URL is never something the user should have to select by hand. Copy button, checkmark confirmation, toast
Connection status at a glance	Colour plus icon plus text on every row. The page header carries an aggregate summary such as "3 of 4 calendars healthy"
Relative plus absolute time	"12 minutes ago" as the primary, exact timestamp on hover. Relative time is scannable, absolute time is what support needs
Show what changed, not just that it ran	The blocked-dates delta is the most useful column in the history table. A sync with zero changes should be visibly a no-op
Progress indicators	Sync Now shows an inline spinner, not a full-page loader. The rest of the page stays usable
Real empty states	Different copy for no connections, no history, and no results after filtering. Each with the right next action
Errors that name the fix	Never "sync failed". Always what happened, what it means and what to do. Pair every error code with a hint string
Inline validation with a live feed test	Validating the feed inside the modal, before saving, converts a class of silent failures into a caught mistake. The event count is the proof of success the user needs
Confirmation for destructive actions only	Delete and regenerate get dialogs. Frequency changes save immediately with a toast and an undo affordance
Manual sync always available	Even on a paused or errored connection. It is how users verify a fix
Sync history retained and filterable	Operators use it to reconstruct what happened around a double booking. Ninety days minimum
Onboarding on first visit	A short two-step guide: paste our link into your channels, then paste their links here. Most operators will not intuit that both directions are needed
Platform-specific help with screenshots	The highest-value content in the feature. Where the export link lives differs on every channel and moves between redesigns
Show the export as connected, not just published	Displaying "last fetched by Airbnb, 22 minutes ago" is the only proof the outbound direction is working. Without it, export is a black hole
Cross-link from the availability calendar	A date blocked by an import should show a chip naming the source, linking to the connection
Do not hide limitations	State on the page that iCal carries availability only, not prices or guest details

18.3 Microcopy reference

Context	Copy
Export section description	Share this link with the channels you sell on so they can block dates that are already booked here.
Export privacy note	Anyone with this link can see which dates are blocked. It contains no guest names or contact details.
Never fetched	No channel has read this feed yet. It can take up to 24 hours after you paste the link.
Import section description	Dates booked on these channels will be blocked here automatically.
Empty state	Connect the channels you sell on and we will block dates here automatically when a guest books somewhere else.
Zero events	This feed is valid but currently empty. That is normal if you have no upcoming bookings on this channel.
Sync in progress	Checking for new bookings...
Conflict alert	A booking on Airbnb overlaps dates already booked here on 14 to 16 August. We kept your booking and did not block those dates.
Delete confirmation	This does not cancel any bookings. It only stops future updates from this channel.
Latency expectation	We publish your changes immediately. Each channel decides how often to read them, usually every 1 to 24 hours.

19. Future Enhancements

19.1 Prioritized roadmap

Phase	Enhancement	Rationale
Fast follow	Google Calendar, Outlook and Apple Calendar as first-class import platforms with tailored help	Operators block personal downtime in their own calendar. This is the most requested addition and requires no new mechanism
Fast follow	Bulk connect across listings	An operator with 40 listings connecting 4 channels each faces 160 manual steps today
Fast follow	Copy sync configuration from one listing to another	Same problem, cheaper solution
Phase 2	Webhook support, outbound	Notify the operator's own systems on booking, block and conflict events. Turns us into a source others can build on
Phase 2	Conflict resolution workspace	A dedicated view for reviewing and resolving conflicts, with a guest-messaging shortcut
Phase 2	Sync health scoring per listing	A single number that surfaces neglected connections before they cause a double booking
Phase 3	Direct OTA APIs for the top channels	Removes the polling delay, adds pricing and inventory, and unlocks guest data. Start with the one or two channels that carry the most volume. Must coexist with iCal via <code>source_type</code>
Phase 3	Real-time push, inbound	Where a channel supports webhooks, replace polling entirely
Phase 3	Seat and capacity inventory sync	Requires an API. This is the structural fix for time-slotted tours
Phase 4	Availability rules engine	Buffer days between bookings, minimum and maximum stay, blackout patterns, applied on top of the imported layer
Phase 4	Pricing synchronization	Rate parity across channels. A separate product, not a calendar feature
Phase 4	Channel manager integrations	Connect to established channel managers instead of building every OTA integration ourselves. Often faster than direct APIs and covers the long tail
Phase 4	Two-way write-back	Push blocks to channels that support it, rather than waiting for them to poll

19.2 Architectural decisions to protect

Three choices in this specification exist specifically so that the roadmap above does not require a rewrite:

- `blocked_dates.source_type` allows iCal imports, direct API channels and manual blocks to coexist per date. Adding a direct API in phase 3 is a new source type, not a migration.
- The import pipeline is a full-state diff rather than an incremental apply, so a channel can be switched from iCal to API and back without corrupting availability.
- Export token settings are stored separately from the listing, so per-channel export variants (a feed that excludes a given source, or a feed with a different horizon) can be added without a schema change.

20. Implementation Notes

20.1 Suggested build sequence

Milestone	Scope	Ships as
M1	Export only. Token model, feed builder, cache, public endpoint, export UI card, fetch logging	Operators can push availability out. Immediately useful on its own
M2	Import, manual only. Add Calendar modal, validation endpoint, fetcher, parser, normalizer, diff engine, blocked dates writer, Sync Now	Operators can pull availability in when they choose
M3	Automation. Scheduler, worker pool, retry ladder, circuit breaker, status model, sync history	The feature becomes real
M4	Reliability and communication. Conflict detector, notifications, error banners, troubleshooting content, platform help articles	Support load drops
M5	Scale. Account-level dashboard, settings, permissions matrix, audit trail, bulk operations	Ready for multi-listing operators

M1 and M2 are independently shippable and independently valuable. Do not hold export behind import.

20.2 Testing requirements

Parser fixture library. Capture real feeds from every supported channel and commit them as fixtures. Include the ugly ones: missing `VERSION`, LF-only line endings, Latin-1 encoding, inclusive `DTEND`, reused `UID`, cancelled events, 3,000-event feeds, feeds with guest names in the summary. Every fixture is a regression test.

Property tests. Syncing the same feed twice produces no changes on the second run. Syncing feed A then feed B then feed A returns to state A. Removing all events releases exactly the dates that were added.

Failure injection. Timeouts, slow-loris responses, 5xx storms, redirect loops, mid-stream connection drops, oversized bodies, TLS errors. Each must produce the correct status, the correct retry behaviour and no partial writes.

Security tests. SSRF attempts against loopback, private ranges, cloud metadata endpoints and DNS rebinding. Token enumeration. Rate limit enforcement. Verification that no guest PII appears in an export feed under any configuration.

Concurrency tests. Two workers claiming the same connection. Delete during an active sync. Booking created during a sync covering the same dates.

Timezone tests. DST transitions in both directions, all-day events across timezone boundaries, listings in half-hour offset zones, feeds with no timezone information at all.

20.3 Non-functional targets

Metric	Target
Export endpoint p95 latency, cache hit	Under 50 ms
Export endpoint p95 latency, cache miss	Under 300 ms
Sync job p95 duration, feed under 100 KB	Under 3 seconds
Scheduler lag, time between due and claimed	Under 60 seconds at p95
Sync success rate excluding user-side URL errors	Above 99 percent
Calendar Sync page time to interactive	Under 1.5 seconds
Maximum sustained sync throughput	10,000 connections per hour per worker pool, horizontally scalable

20.4 Open questions for the product owner

1. **Plan gating.** Which sync intervals sit behind which tier, and what is the calendar cap per listing per tier?
2. **Time-slotted tours.** Do we ship iCal for slot-based listings with a clear warning, or block it until API-based slot sync exists? Recommendation is to ship with a warning, since day-level blocking is still better than nothing.
3. **Export horizon.** Is past 30 days plus 24 months correct for the inventory mix, or do multi-year advance bookings need a longer window?
4. **Conflict default.** Confirm that `keep_local` is right. It assumes our platform's booking always wins, which is correct if we are the source of truth but worth an explicit decision.
5. **Retention.** Ninety days of sync history on standard plans. Confirm against support needs, since double-booking disputes can surface months later.
6. **Which channels get first-class help content at launch?** The screenshot-level guides are expensive to maintain. Recommendation is the six named in the brief, plus Google Calendar.